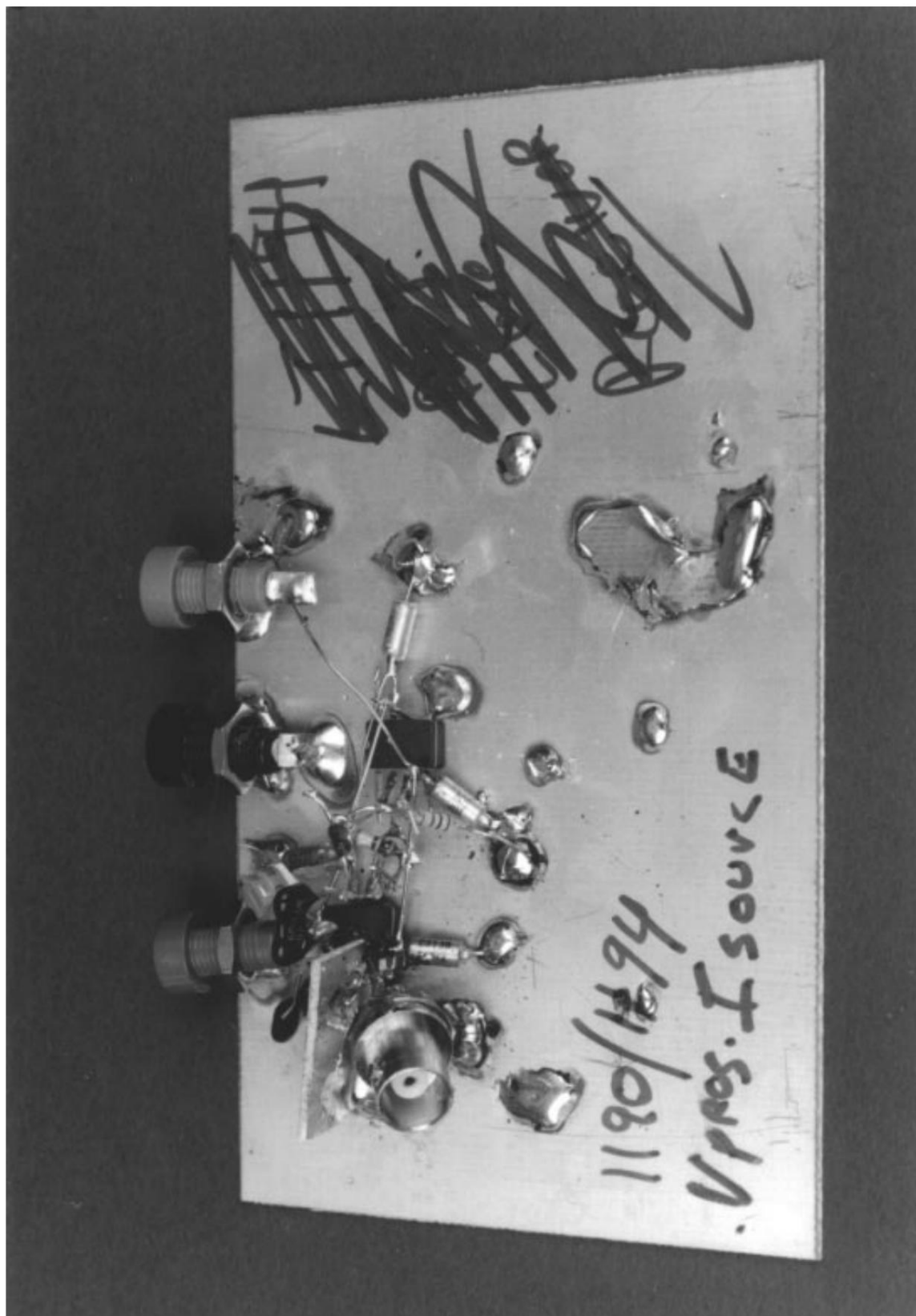
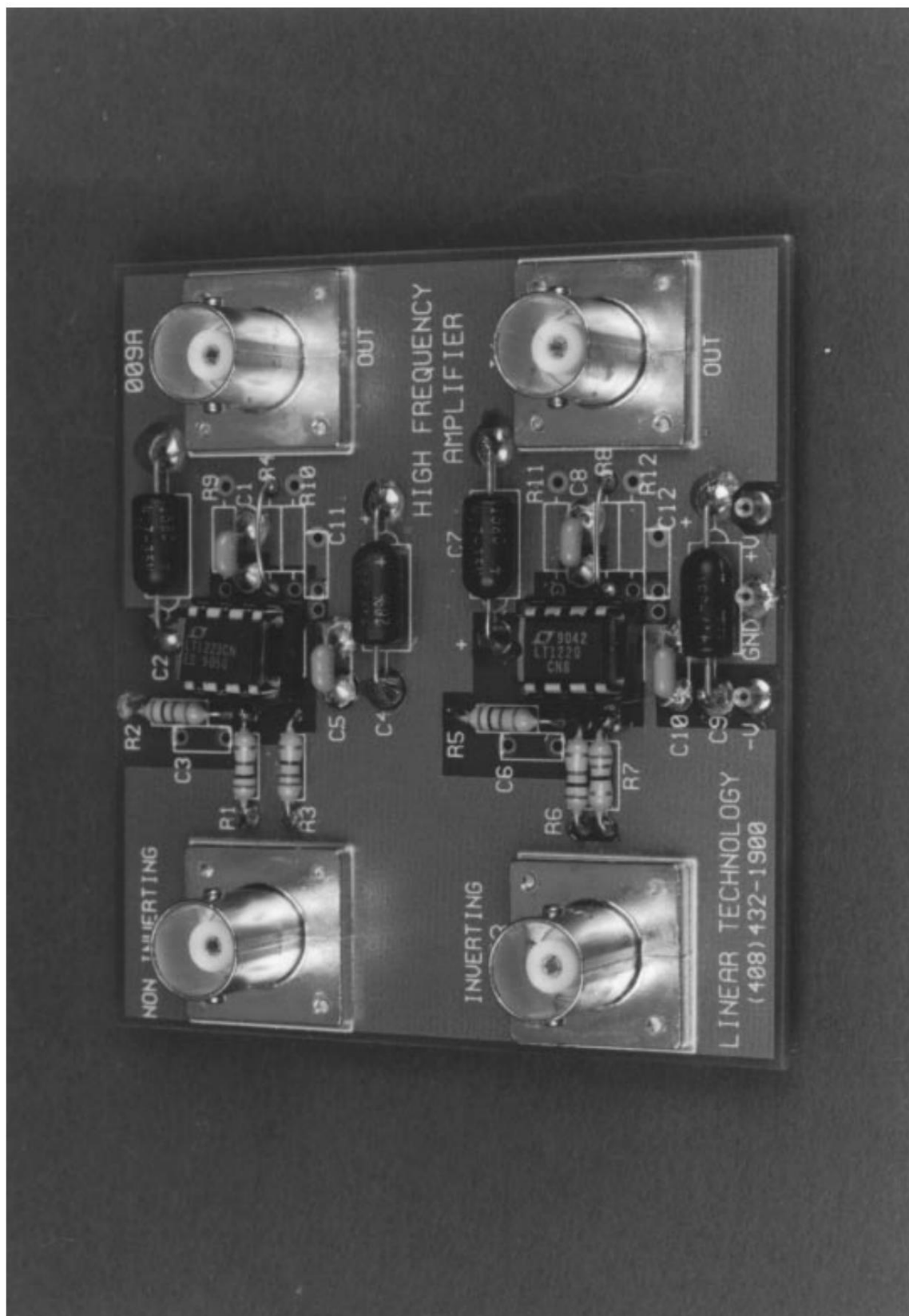




应用笔记 47



附录 G

商业 AM 广播电台的 FCC 许可和施工许可证申请

根据图116电路的应用,以及我们守法的本性,找到近似的仿制品

Federal Communications Commission
Washington, D. C. 20554

Approved by OMB
3000-0037
Expires 2/29/92
See Page 25 for information
regarding public burden estimate

FCC 301

APPLICATION FOR CONSTRUCTION PERMIT FOR COMMERCIAL BROADCAST STATION

For COMMISSION Fee Use Only

FEE NO:
FEE TYPE:
FEE AMT:
ID SEQ:

For APPLICANT Fee Use Only
Is a fee submitted with this application? ☐ Yes ☐ No
If fee exempt (see 47 C.F.R. Section 1.1112), indicate reason therefor (check one box):
☐ Noncommercial educational licensee
☐ Governmental entity
FOR COMMISSION USE ONLY
FILE NO.

Section I - GENERAL INFORMATION

1. Name of Applicant
Street Address or P.O. Box
City State ZIP Code
Telephone No. (include area code)

Send notices and communications to the following person at the address below:
Name
Street Address or P.O. Box
City State ZIP Code
Telephone No. (include area code)

2. This application is for: ☐ AM ☐ FM ☐ TV

(a) Channel No. or Frequency
(b) Principal Community City State

(c) Check one of the following boxes:
☐ Application for NEW station
☐ MAJOR change in licensed facilities; call sign: _____
☐ MINOR change in licensed facilities; call sign: _____
☐ MAJOR modification of construction permit; call sign: _____
File No. of construction permit: _____
☐ MINOR modification of construction permit; call sign: _____
File No. of construction permit: _____
☐ AMENDMENT to pending application; Application file number: _____

NOTE: It is not necessary to use this form to amend a previously filed application. Should you do so, however, please submit only Section I and those other portions of the form that contain the amended information.

3. Is this application mutually exclusive with a renewal application? ☐ Yes ☐ No
If Yes, state: Call letters Community of License City State

图G1

下面是专用的 FCC 申请。完整的表格可写信至：

联邦通信委员会 华盛顿特区 20554

Federal Communications Commission
Washington, D. C. 20554

Approved by OMB
3000-0038
Expires 9/26/90

APPLICATION FOR NEW BROADCAST STATION LICENSE
(Carefully read instructions before filing out Form)
RETURN ONLY FORM TO FCC

For Commission Fee Use Only

FEE NO:
FEE TYPE:
FEE AMT:
ID SEQ:

For APPLICANT Fee Use Only
Is a fee submitted with this application? ☐ Yes ☐ No
If No, indicate reason therefor (check one box):
☐ Nonfeeable application
Fee Exempt (See 47 C.F.R. Section 1.1112)
☐ Noncommercial educational licensee
☐ Governmental entity

SECTION I - GENERAL DATA

Legal Name of Applicant
Mailing Address
City State Zip Code
Telephone No. (include area code)

1. Facilities authorized by construction permit
This application is for: ☐ Commercial ☐ Noncommercial
☐ AM Directional ☐ AM Non-Directional ☐ FM Directional ☐ FM Non-Directional ☐ TV

Call Letters Community of License Construction Permit File No. Modification of Construction Permit File No(s) Expiration Date of Last Construction Permit

2. Is the station now operating pursuant to automatic program test authority in accordance with 47 C.F.R. Section 73.1620? ☐ Yes ☐ No
If No, explain.

3. Have all the terms, conditions, and obligations set forth in the above described construction permit been fully met? ☐ Yes ☐ No
If No, state exceptions.

4. Apart from the changes already reported, has any cause or circumstance arisen since the grant of the underlying construction permit which would result in any statement or representation contained in the construction permit application to be now incorrect? ☐ Yes ☐ No
If Yes, explain.

5. Has the permittee filed its Ownership Report (FCC Form 323) or ownership certification in accordance with 47 C.F.R. Section 73.3615(b)? ☐ Yes ☐ No
If No, explain.

图G2

图G1-G2.适用于图 116 电路的 FCC 表格

应用笔记 47

附录H

关于当前反馈

与一些热情的营销主张相反,目前的反馈并不新鲜。事实上,它比“正常”电压反馈要古老得多,而“正常”电压反馈在运算放大器中已经非常普及。当前的反馈连接已有 50 年历史,甚至可能更老。至少 William R. Hewlett 在 1939 年使用它构建了他现在著名的正弦波

振荡器。1 “阴极反馈”在整个 30 年代、40 年代和 50 年代广泛应用于射频和宽带仪器设计。这是一种最喜欢的反馈形式,如果没有其他原因,那就是没有其他地方可以反馈了!

1950 年代初期,GA Philbrick Researches 推出了 K2-W,这是第一款商用封装运算放大器。该器件具有高阻抗差分输入,允许当今常见的电压型反馈。尽管低频仪器类型很快就利用了高阻抗反馈节点所提供的增加的效用,但射频和宽带设计人员几乎没有注意到。他们继续使用阴极反馈,在新的晶体管形式中称为(还有什么?)发射极反馈。

在射频和宽带仪器中继续使用电流反馈的许多例子可以在

设计可追溯到 1950 年代至今。2 电压型反馈在这一时期已成为现实,表面上更容易使用,特别是随着单片器件变得更便宜,为什么还要继续使用分立电流反馈?当前技术持续流行的原因是(并且仍然是)带宽。

当前的反馈速度要快得多。此外,在限制范围内,基于电流反馈的放大器的带宽不会随着闭环增益的增加而降低。

与电压反馈放大器相比,这是一个显著的优势,电压反馈放大器的带宽随着闭环增益的增加而下降。

最近,基于电流的设计已成为通用、易于使用的单片和混合设备。这为更广泛的受众带来了高速功能,有望开辟新的应用程序。因此,虽然这项技术并不新鲜,尽管营销声称,但机会却是。尽管基于电流的设计的直流性能比电压放大器差,但其带宽优势是不可否认的。有什么魔力?

电流反馈基础知识

威廉·格罗斯

电流反馈放大器与电压反馈放大器的区别一开始并不明显,因为从外部来看,差异可能很微妙。

两种放大器类型使用相似的符号,并且可以使用相同的方程在一阶基础上应用。

然而,它们在增益带宽权衡和大信号响应方面的行为则是另一回事。

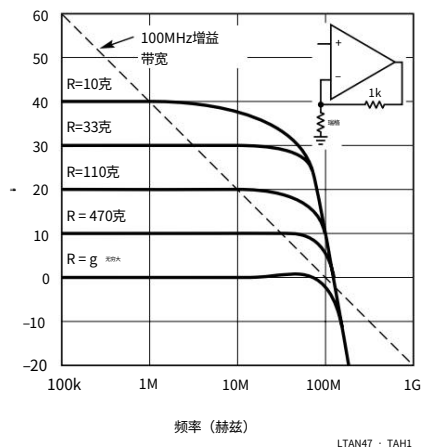
与电压反馈放大器不同,电流反馈放大器中的小信号带宽不是闭环增益的直接反函数,并且大信号响应更接近理想。这两个好处都是因为反馈电阻器决定了驱动放大器内部补偿电容器的电流。事实上,放大器从输出到反相输入的反馈电阻(R_f)与内部结电容一起设置闭环带宽。尽管从反相输入到接地的增益设置电阻器(R_g)与 R_f 一起设置电压增益,就像在电压反馈运算放大器中一样,闭环带宽也不会改变。对此的解释相当简单。CFA 的等效增益带宽积由反相输入处的戴维宁等效电阻和内部补偿电容器设置。如果 R_f 保持恒定并且增益随 R_g 变化,则戴维南电阻的变化量与增益相同。从整体环路的角度来看,反馈衰减的这种变化将导致噪声增益的变化,以及开环带宽的成比例减少(如在传统运算放大器中)。然而,对于电流反馈,关键是戴维南电阻的变化也会产生补偿性变化

注 1:参见参考文献 19 中的附录 C,“维恩桥和休利特先生”。另请参见参考文献 20 至 24 和 46。

注 2:请参阅 GE 于 1964 年出版的“通用电气晶体管手册”。另请参阅 Hewlett-Packard 3400A RMS 电压表、1120A FET 探头和 Tektronix P6042 电流探头的操作和服务手册。

与传统的固定增益带宽放大器不同,它具有开环带宽。因此,电流反馈放大器的净闭环带宽对于各种闭环增益保持相同。

图 H1 显示了驱动 100Ω 的五种增益设置下 LT1223 电压增益与频率的关系。为了进行比较,显示了电压反馈放大器具有的固定 100MHz 增益带宽限制的曲线图。显然,对于大于 1 的增益,LT1223 可提供 3-20 倍的带宽。



图H1.电流反馈放大器 (曲线族)和传统电压放大器 (直线)的电压增益与频率的关系

由于反馈电阻器决定了 LT1223 的补偿,因此几乎可以针对每种应用来优化带宽和瞬态响应。当在 $\pm 15\text{V}$ 电源上工作时, R_f 应为 $1\text{k}\Omega$ 或更大以保持稳定性,但在 $\pm 5\text{V}$ 电源上工作时,最小值为 680Ω ,因为结电容随电压降低而增大。对于任一情况,也可以使用更大的反馈电阻,但会减慢 LT1223 的速度 (这在某些应用中可能是理想的)。

LT1223 提供出色的转换速率和带宽,以及比以前的电流馈送更好的 DC 性能

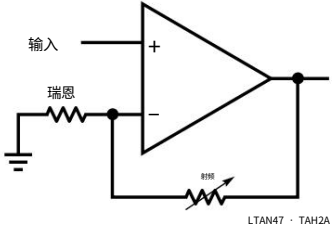
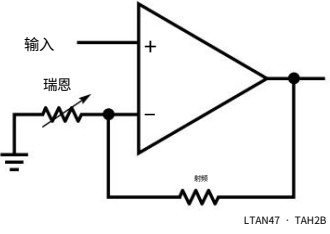
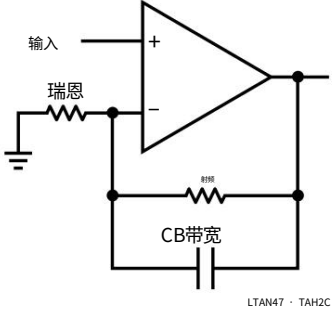
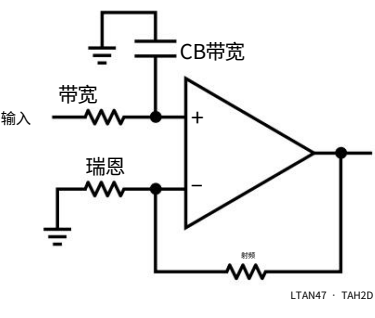
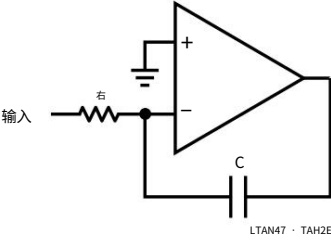
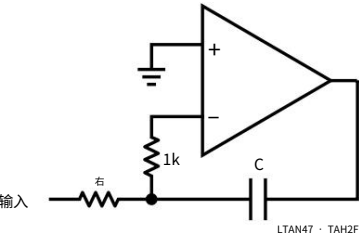
后置放大器 (CFA)。在具有 1k 反馈电阻的 $\pm 15\text{V}$ 电源上,小信号带宽在 400Ω 负载中为 100MHz ,在 100Ω 负载中为 75MHz 。输入将遵循 $250\text{V}/\mu\text{s}$ 的压摆率,输出产生超过 $500\text{V}/\mu\text{s}$,并且输出压摆率远超过 $1000\text{V}/\mu\text{s}$,以实现大输入过驱动。输入失调电压为 3mV (最大值),输入偏置电流为 $3\mu\text{A}$ (最大值)。一个 $10\text{k}\Omega$ 电位器连接到引脚 1 和 5,并将抽头连接到 $V+$,提供可选的偏移微调。这种调整可将反相输入电流偏移约 $\pm 10\mu\text{A}$,从而有效地产生输入电压偏移。

LT1223 还具有通过引脚 8 提供的关断控制功能。从引脚 8 拉动超过 $200\mu\text{A}$ 的电流会将电源电流降至 3mA 以下,并使输出进入高阻抗状态。强制关闭的简单方法是使用集电极开路 (漏极) 逻辑级将引脚 8 接地。

内部电阻器限制电流,允许直接连接,无需额外部件。当引脚 8 开路时,LT1223 正常工作。

运算放大器和 CFA 之间工作特性的差异导致常见电路配置略有差异。图 H2 总结了一些流行的电路类型,显示了运算放大器和 CFA 之间的差异。增益可以通过运算放大器中的 R_{IN} 或 R_f 设置,而 CFA 的反馈电阻 (R_f) 是固定的。运算放大器带宽可通过反馈电容器进行控制;对于 CFA,输入带宽必须受到限制。从未使用反馈电容器。在积分器中, 1k 电阻器必须包含在 CFA 中,以便其负输入具有最佳阻抗。最后, (未显示) CFA 输入的偏置电流之间不存在相关性。因此,源阻抗匹配不会提高直流精度。匹配输入源阻抗有助于提高没有内部偏置电流消除功能的运算放大器的偏置性能。

应用笔记 47

电路类型	电压反馈 放大器	当前反馈 功放
设置增益	 LTAN47 · TAH2A	 LTAN47 · TAH2B
限制带宽	 LTAN47 · TAH2C	 LTAN47 · TAH2D
积分器	 LTAN47 · TAH2E	 LTAN47 · TAH2F

图H2。应用电流放大器和电压放大器的一些实际差异

附录一

高频放大器评估板

LTC 演示板 009 (照片,图 I1,原理图,图 I2)旨在简化高速运算放大器的评估。它包括反相和非反相电路,并提供额外的孔以允许使用板装 BNC 或 SMA 连接器。除了共享电源和接地连接之外,这两个电路是独立的。

布局是影响任何高速放大器性能的主要因素。不良的布局技术会对成品电路的性能产生不利影响。演示板 009 中全部使用的几种重要布局技术描述如下:

1. 顶部接地层。接地层的主要任务是降低接地连接的阻抗。

均匀铜片上任意两点之间的电感小于连接相同两点的薄而直的铜迹的电感。

接地层近似铜片的特性,并降低电路中关键点的阻抗,例如连接器和电源旁路电容器的接地。

2. 接地层空隙。某些元件和电路节点对杂散电容非常敏感。两个很好的例子是运算放大器的求和节点和反馈电阻。在这些区域的接地层中留有空隙,以减少杂散接地电容。

3. 输入/输出匹配。输入和输出走线的宽度调整为 50Ω 的带状线阻抗。请注意,终端电阻 ($R3$ 和 $R7$) 连接到输入线的末端,而不是连接器处。虽然带状线技术对于演示板来说并不是绝对必要的,但它们对于线路长度较长的较大布局非常重要。演示板上的短线可以以 50Ω 、 75Ω 或 93Ω 端接,而不会对性能产生不利影响。4. 输入和输出地的分离。即使接地层呈现低阻抗,输入和输出接地仍然是分开的。例如,终端电阻 ($R3$ 和 $R7$)和增益设置电阻 ($R1$)在输入连接器附近接地。电源旁路电容器 ($C1$ 、 $C2$ 、 $C4$ 、 $C5$ 、 $C7$ 、 $C8$ 、 $C9$ 和 $C10$) 在输出连接器附近返回接地。

该电路板设计用于容纳标准 8 引脚 miniDIP、单运算放大器,例如 LT1190 和 LT1220 系列。电压和电流反馈类型均可使用。引脚 1、5 和 8 配备了额外的孔,用于调整 DC 偏移、补偿,或者对于 LT1223 和 LT1190/1/2,用于关闭放大器。

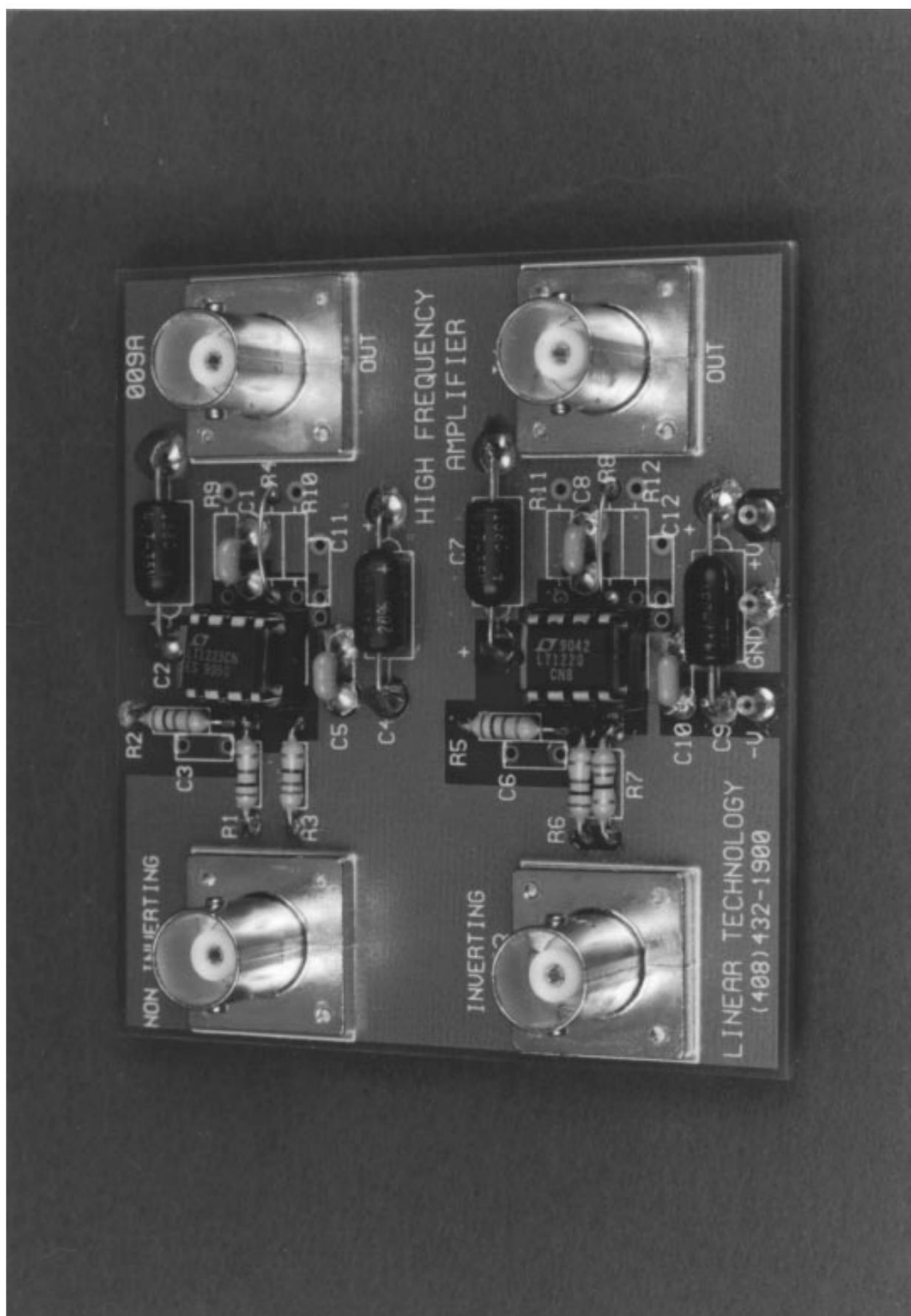
如果正在评估 LT1223 等电流反馈放大器,则省略 $C3$ / $C6$ 。 $R4$ 和 $R6$ 用于驱动低阻抗线路时的阻抗匹配。如果放大器直接驱动线路,或者负载阻抗较高,可以用跳线代替 $R4$ 和 $R8$ 。

类似地, $R10$ 和 $R12$ 可用于在放大器的输出端建立负载。

运算放大器可使用薄型插座以方便更换部件,但在 100MHz 以上时性能可能会受到影响。

当高速运算放大器的电源引脚被射频品质电容器旁路时,其工作效果最佳。 $C1$ 、 $C5$ 、 $C8$ 和 $C10$ 应为 10nF 圆盘陶瓷或其他自谐振频率大于 10MHz 的电容器。极化电容器 ($C2$ 、 $C4$ 、 $C7$ 和 $C9$) 应为 1 μ F 至 10 μ F 钽电容器。大多数 10nF 陶瓷在远高于 10MHz 时自谐振,4.7 μ F 固体钽 (轴向引线) 在 1MHz 或以下时自谐振。引线长度至关重要;通过 2 引线测量时,4.7 μ F 钽的自谐振频率下降了 2 倍。尽管电容器在高频下可能会变得感性,但由于阻抗较低,它仍然是高于谐振的有效旁路元件。

应用笔记 47



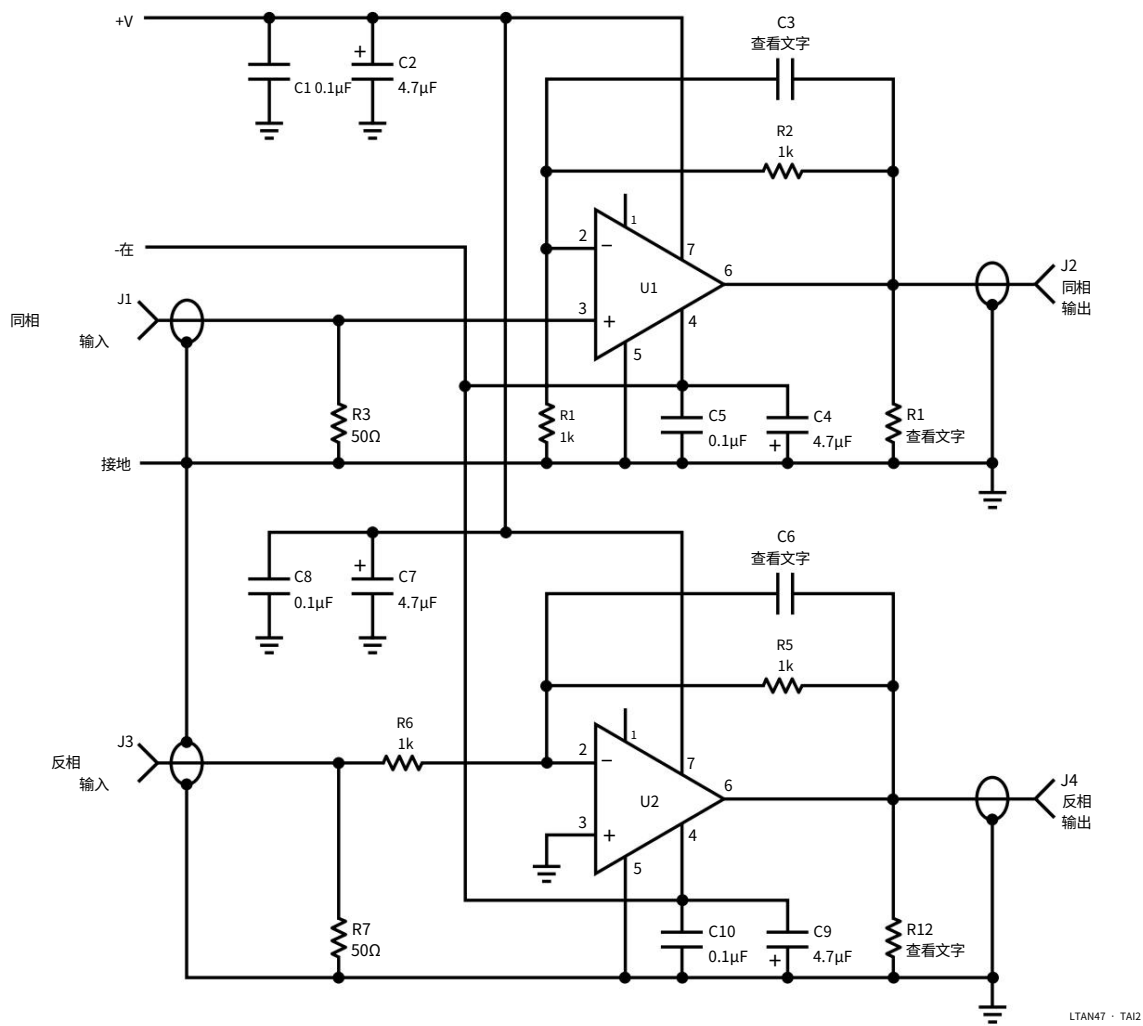


图 12.高频放大器演示板原理图

应用笔记 47

APPENDIX J

Reprinted, with permission from Cahners Publishing Co., from *EEE Magazine*, Vol. 15, No. 8, August 1967.

The Contributions of Edsel Murphy to the Understanding of the Behavior of Inanimate Objects

D. L. KLIPSTEIN

Abstract—Consideration is given to the effects of the contributions of Edsel Murphy to the discipline of electronics engineering. His law is stated in both general and special form. Examples are presented to corroborate the author's thesis that the law is universally applicable.

I. INTRODUCTION

IT HAS LONG BEEN the consideration of the author that the contributions of Edsel Murphy, specifically his general and special laws delineating the behavior of inanimate objects, have not been fully appreciated. It is deemed that this is, in large part, due to the inherent simplicity of the law itself.

It is the intent of the author to show, by references drawn from the literature, that the law of Murphy has produced numerous corollaries. It is hoped that by noting these examples, the reader may obtain a greater appreciation of Edsel Murphy, his law, and its ramifications in engineering and science.

As is well known to those versed in the state-of-the-art, Murphy's Law states that "If anything can go wrong, it will." Or, to state it in more exact mathematical form:

$$1 + 1 \approx 2 \quad (1)$$

where $\approx$ is the mathematical symbol for hardly ever.

Some authorities have held that Murphy's Law was first expounded by H. Cohen¹ when he stated that "If anything can go wrong, it will — during the demonstration." However, Cohen has made it clear that the broader scope of Murphy's general law obviously takes precedence.

To show the all-pervasive nature of Murphy's work, the author offers a small sample of the application of the law in electronics engineering.

II. GENERAL ENGINEERING

II.1. A patent application will be preceded by one week by a similar application made by an independent worker.

II.2. The more innocuous a design change appears, the further its influence will extend.

II.3. All warranty and guarantee clauses become void upon payment of invoice.

II.4. The necessity of making a major design change

Manuscript received April 17, 1967; revised June 3, 1967. The work reported herein has not been supported by grants from the Central Intelligence Agency.

The author is Director of Engineering at Measurement Control Devices, 2445 Emerald Street, Philadelphia, Pa.

increases as the fabrication of the system approaches completion.

II.5. Firmness of delivery dates is inversely proportional to the tightness of the schedule.

II.6. Dimensions will always be expressed in the least usable term. Velocity, for example, will be expressed in furlongs per fortnight.²

II.7. An important Instruction Manual or Operating Manual will have been discarded by the Receiving Department.

II.8. Suggestions made by the Value Analysis group will increase costs and reduce capabilities.

II.9. Original drawings will be mangled by the copying machine.³

III. MATHEMATICS

III.1. In any given miscalculation, the fault will never be placed if more than one person is involved.

III.2. Any error that can creep in, will. It will be in the direction that will do the most damage to the calculation.

III.3. All constants are variables.

III.4. In any given computation, the figure that is most obviously correct will be the source of error.

III.5. A decimal will always be misplaced.

III.6. In a complex calculation, one factor from the numerator will always move into the denominator.

IV. PROTOTYPING AND PRODUCTION

IV.1. Any wire cut to length will be too short.

IV.2. Tolerances will accumulate unidirectionally toward maximum difficulty of assembly.

IV.3. Identical units tested under identical conditions will not be identical in the field.

IV.4. The availability of a component is inversely proportional to the need for that component.

IV.5. If a project requires n components, there will be $n-1$ units in stock.⁴

IV.6. If a particular resistance is needed, that value will not be available. Further, it cannot be developed with any available series or parallel combination.⁵

IV.7. A dropped tool will land where it can do the most damage. (Also known as the law of selective gravitation.)

IV.8. A device selected at random from a group having 99% reliability, will be a member of the 1% group.

IV.9. When one connects a 3-phase line, the phase

sequence will be wrong.⁶

IV.10. A motor will rotate in the wrong direction.⁷

IV.11. The probability of a dimension being omitted from a plan or drawing is directly proportional to its importance.

IV.12. Interchangeable parts won't.

IV.13. Probability of failure of a component, assembly, subsystem or system is inversely proportional to ease of repair or replacement.

IV.14. If a prototype functions perfectly, subsequent production units will malfunction.

IV.15. Components that must not and cannot be assembled improperly will be.

IV.16. A dc meter will be used on an overly sensitive range and will be wired in backwards.⁸

IV.17. The most delicate component will drop.⁹

IV.18. Graphic recorders will deposit more ink on humans than on paper.¹⁰

IV.19. If a circuit cannot fail, it will.¹¹

IV.20. A fail-safe circuit will destroy others.¹²

IV.21. An instantaneous power-supply crowbar circuit will operate too late.¹³

IV.22. A transistor protected by a fast-acting fuse will protect the fuse by blowing first.¹⁴

IV.23. A self-starting oscillator won't.

IV.24. A crystal oscillator will oscillate at the wrong frequency — if it oscillates.

IV.25. A pnp transistor will be an npn.¹⁵

IV.26. A zero-temperature-coefficient capacitor used in a critical circuit will have a TC of $-750 \text{ ppm}/^\circ\text{C}$.

IV.27. A failure will not appear till a unit has passed Final Inspection.¹⁶

IV.28. A purchased component or instrument will meet its specs long enough, and only long enough, to pass Incoming Inspection.¹⁷

IV.29. If an obviously defective component is replaced in an instrument with an intermittent fault, the fault will reappear after the instrument is returned to service.¹⁸

IV.30. After the last of 16 mounting screws has been removed from an access cover, it will be discovered that the wrong access cover has been removed.¹⁹

IV.31. After an access cover has been secured by 16 hold-down screws, it will be discovered that the gasket has been omitted.²⁰

IV.32. After an instrument has been fully assembled, extra components will be found on the bench.

IV.33. Hermetic seals will leak.

V. SPECIFYING

V.1. Specified environmental conditions will always be exceeded.

V.2. Any safety factor set as a result of practical experience will be exceeded.

V.3. Manufacturers' spec sheets will be incorrect by a factor of 0.5 or 2.0, depending on which multiplier gives the most optimistic value. For salesmen's claims these factors will be 0.1 or 10.0.

V.4. In an instrument or device characterized by a number of plus-or-minus errors, the total error will be the sum of all errors adding in the same direction.

V.5. In any given price estimate, cost of equipment will exceed estimate by a factor of 3.²¹

V.6. In specifications, Murphy's Law supersedes Ohm's.

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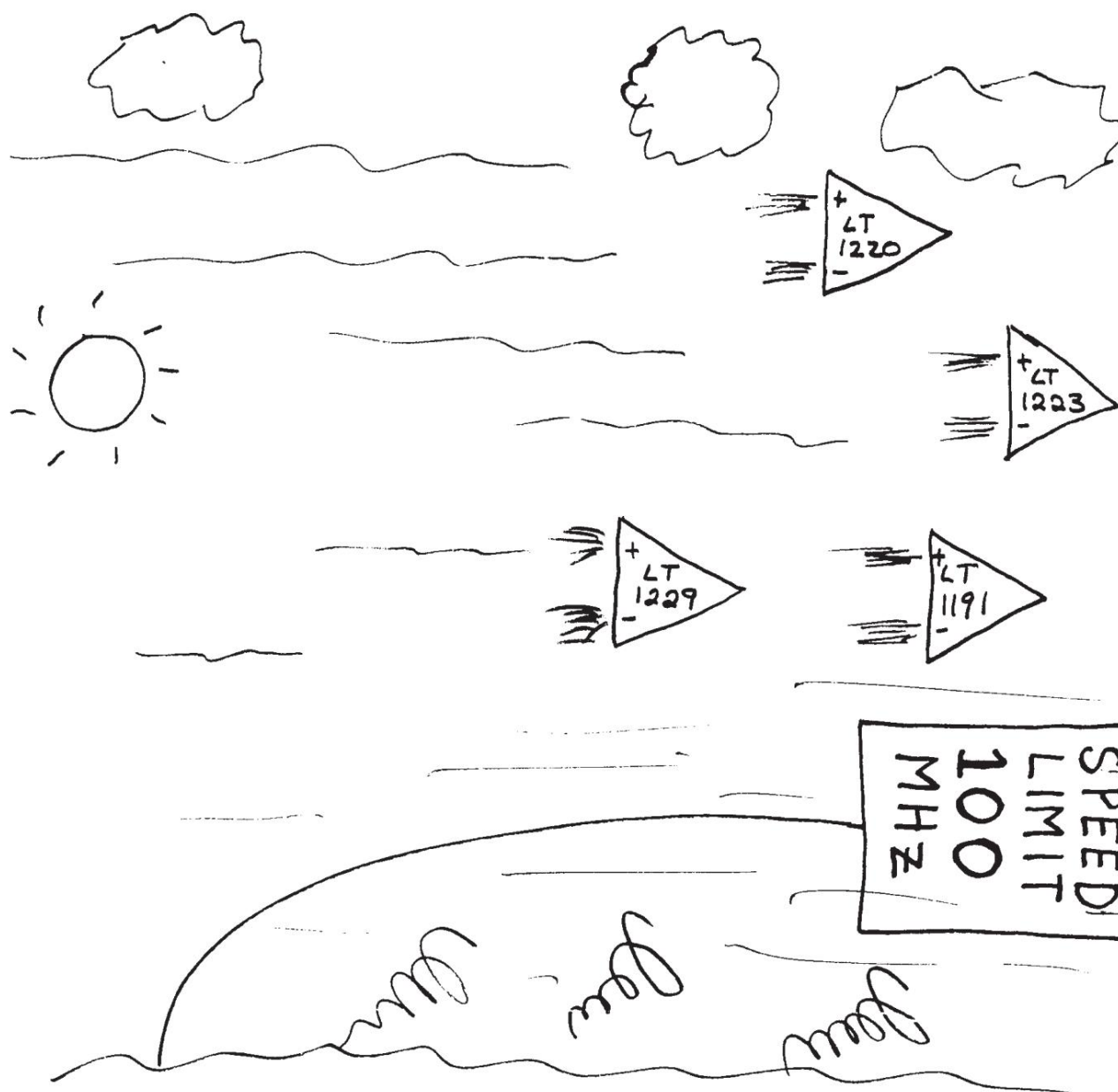
*In some cases where no reference is given, the source material was misplaced during preparation of this paper (another example of Murphy's Law). In accordance with the law, these misplaced documents will turn up on the date of publication of this paper.



The man who developed one of the most profound concepts of the twentieth century is practically unknown to most engineers. He is a victim of his own law. Destined for a secure place in the engineering hall of fame, something went wrong.

His real contribution lay not merely in the discovery of the law but more in its universality and in its impact. The law itself, though inherently simple, has formed a foundation on which future generations will build.

应用笔记 47



(I WISH This App Note WENT AS
FAST AS LTC Amplifiers)

Willie
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